Nephrotic Syndrome Type 18 (NPHS18 / NUP133-related Steroid-Resistant Nephrotic Syndrome): A Comprehensive Disease Characteristics Report

Disease: Nephrotic Syndrome Type 18 (NPHS18) **Causal gene: NUP133** (Nucleoporin 133 kDa; HGNC:29915; NCBI Gene 55746; UniProt Q8WUM0) **OMIM:** 618177 · **MONDO:** MONDO:0032581 · **Category:** Mendelian, autosomal recessive

Summary

Nephrotic Syndrome Type 18 (NPHS18) is a rare autosomal-recessive, steroid-resistant nephrotic syndrome (SRNS) caused by biallelic (homozygous or compound-heterozygous) hypomorphic variants in *NUP133*, the gene encoding nucleoporin 133 kDa — a scaffold component of the outer-ring Nup84/Y-complex of the nuclear pore complex (NPC). The disease was defined when Braun et al. (2018) identified recessive mutations in four outer-ring nucleoporin genes — *NUP107*, *NUP85*, *NUP133*, and *NUP160* — across 13 families with SRNS (PMID: 30179222). Despite NUP133 being a ubiquitously expressed, essential housekeeping protein, its disease manifestations are strikingly tissue-restricted, predominantly affecting the kidney's glomerular podocytes and, in syndromic forms, the developing brain.

The core mechanistic insight is that loss of NUP133 function activates **CDC42**, a Rho-family GTPase and a key effector of SRNS pathogenesis, in podocytes. This leads to foot-process effacement, focal segmental glomerulosclerosis (FSGS) — seen in ~89% of nucleoporin-SRNS biopsies — and progression to kidney failure at a median age of ~13 years for NUP133. The phenotype spans a continuum from **kidney-limited SRNS** (Braun 2018) to syndromic **Galloway-Mowat syndrome (GAMOS)** featuring microcephaly, brain malformation, and early-onset nephrotic syndrome (Fujita 2018, PMID: 30427554). Why a ubiquitous scaffold nucleoporin causes selective podocyte and neuronal disease is explained by cell-type-specific

differentiation dependence: Nup133 is dispensable for pluripotency but required for neuroectodermal/neuronal differentiation and for maintaining the highly specialized, non-regenerating podocyte.

Clinically, NUP133-SRNS is **resistant to immunosuppression** (steroids, calcineurin inhibitors, mycophenolate, rituximab). Management is therefore supportive and nephroprotective (RAAS blockade, diuretics, statins, thromboprophylaxis), with **kidney transplantation** as definitive therapy for kidney failure — favored because podocyte-intrinsic genetic SRNS has low post-transplant recurrence. The single greatest genotype-guided benefit of early genetic diagnosis is avoiding prolonged, toxic, ineffective immunosuppression. This report synthesizes 12 confirmed findings across 43 reviewed papers into a complete disease knowledge base entry.

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1. Disease Information

Overview. NPHS18 is a monogenic (Mendelian) form of steroid-resistant nephrotic syndrome. Nephrotic syndrome is clinically defined by nephrotic-range proteinuria (>3.5 g/day in adults; >40 mg/m²/h in children), hypoalbuminemia (<2.5–3 g/dL), peripheral edema, and hyperlipidemia. In NPHS18 the glomerular lesion is characteristically **focal segmental glomerulosclerosis (FSGS)** with podocyte foot-process effacement on electron microscopy. The disease belongs to the broader class of **nucleoporin-associated SRNS ("nucleoporopathies")**, caused by mutations in genes encoding the NPC. *NUP133* sits on the same disease axis as its physical binding partner *NUP107* and the other outer-ring nucleoporin genes (*NUP107*, *NUP85*, *NUP160*) and inner-ring genes (*NUP93*, *NUP205*).

(This disease serial number refers specifically to NUP133 — not to be confused with other NPHS numbers mapping to NPHS1/NPHS2/PLCE1, etc.)

Key identifiers.

Resource	Identifier
OMIM	618177
MONDO	MONDO:0032581
Gene (HGNC)	NUP133 / HGNC:29915
NCBI Gene	55746
UniProt	Q8WUM0
Ensembl	ENSG00000069248 (chr1q42.12)
Reference transcript	NM_018230.3 (protein 1156 aa)

Synonyms / alternative names: NPHS18; NUP133-related steroid-resistant nephrotic syndrome; NUP133-related nucleoporopathy; (syndromic form) Galloway-Mowat syndrome due to NUP133.

Information source. Disease-level characterization derives from aggregated resources (OMIM, Orphanet, ClinVar, gnomAD, UniProt) and from primary case reports/series of individual patients (Braun 2018; Fujita 2018; reviews such as Yao 2025, PMID: 39331077). It is a rare disease known from a modest number of families worldwide; no large EHR-based registry specific to NUP133 exists.

2. Etiology

Primary cause — genetic. NPHS18 is caused exclusively by **biallelic loss-of-function/hypomorphic variants in *NUP133***. Inheritance is autosomal recessive; both alleles must be affected. Reported disease alleles include splice-site, missense, and frameshift variants (Section 4). There is no environmental or infectious cause; the nephrotic syndrome is a direct consequence of the podocyte-intrinsic genetic defect.

Genetic risk factors. The sole risk factor is carriage of two pathogenic *NUP133* alleles. **Consanguinity** substantially enriches recessive SRNS, as biallelic (often homozygous) variants are more likely in offspring of related parents (Azouaou 2026 Algerian consanguineous cohort,

PMID: 41612037; Soliman 2025, PMID: 40402239). Modifier genes within the Y-complex (e.g., *NUP107*) are plausible contributors to expressivity because NUP133 function depends on NUP107 binding, but specific modifier alleles are not established.

Environmental risk / protective factors. None identified. Because this is a Mendelian podocytopathy, classical environmental risk factors (toxins, lifestyle, infections) and protective factors (diet, exposures) are **not applicable** to disease causation. Supportive measures (sodium restriction, nephroprotection) modify disease *course*, not disease *occurrence*.

Gene–environment interactions. No specific GxE interactions are documented. Disease penetrance in biallelic carriers appears high; environmental modulation of severity has not been demonstrated.

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3. Phenotypes

NPHS18 presents a spectrum from isolated (kidney-limited) SRNS to syndromic GAMOS. Core and extrarenal phenotypes with suggested HPO terms:

Phenotype	Type	HPO term	Onset / severity / frequency
Nephrotic syndrome	Clinical/lab	HP:0000100	Childhood (median ~9 y); severe; near-universal
Proteinuria (nephrotic-range)	Lab	HP:0000090 / HP:0003076	Childhood; progressive; universal
Hypoalbuminemia	Lab	HP:0003073	Concurrent with onset; frequent
Edema	Clinical sign	HP:0000969	Presenting feature; frequent
Hyperlipidemia	Lab	HP:0003077	Concurrent; frequent
Focal segmental glomerulosclerosis	Pathology	HP:0000097	~89% of nucleoporin-SRNS biopsies
Podocyte foot-process effacement	Pathology (EM)	—	Characteristic; near-universal
Stage 5 CKD / kidney failure	Clinical	HP:0003774 / HP:0000083	Median ~13 y (NUP133); progressive
Microcephaly (GAMOS)	Physical	HP:0000252	Congenital; syndromic subset
Brain atrophy/anomalies (GAMOS)	Imaging/ structural	HP:0012444	Syndromic subset
Developmental delay / intellectual disability	Behavioral/ neuro	HP:0001263 / HP:0001249	Syndromic subset
Hypotonia	Neuro	HP:0001252	Syndromic subset
Venous thrombosis (NS complication)	Clinical	HP:0004936	Secondary complication
Increased infection susceptibility	Clinical	HP:0002719	Secondary complication

Characteristics. Onset is typically **childhood** (median ~9 years for NUP133; Yao 2025, PMID: 39331077). The course is **steroid-resistant and progressive**, advancing to kidney failure. Extrarenal manifestations of nucleoporin-SRNS include "neurological disorders, growth and development disorders, cardiovascular disorders, and congenital malformations" (Yao 2025). In the syndromic (GAMOS) form, Fujita et al. documented microcephaly, brain anomalies including focal cortical dysplasia type IIa on autopsy, and early-onset nephrotic syndrome with FSGS (PMID: 30427554).

Quality-of-life impact. Progression to dialysis/transplantation and (in GAMOS) neurodevelopmental impairment substantially reduce quality of life. Disease-specific QoL instruments have not been applied to this rare disorder; impact is inferred from the severity of pediatric kidney failure and neurodisability. Secondary complications (thrombosis, infection) from the hypercoagulable, immunocompromised nephrotic state add further morbidity (Ponticelli 2023, PMID: 37466816).

4. Genetic / Molecular Information

Causal gene. *NUP133* (HGNC:29915; disease OMIM 618177), chr1q42.12, ENSG00000069248; reference transcript NM_018230.3 encoding a 1156-amino-acid protein (UniProt Q8WUM0).

Pathogenic variants. Disease-causing alleles are rare and biallelic. Key reported variants:

Variant (NM_018230.3)	Type	Phenotype	Functional effect	Source
c.3335-11T>A (9-bp intronic insertion, exons 25–26)	Splice-site	GAMOS (syndromic)	Impairs NUP133–NUP107 interaction	Fujita 2018 PMID: 30427554
Compound-het / homozygous missense & splice alleles	Missense/ splice	SRNS (kidney-limited)	Weaken neighboring NPC-subunit interactions	Braun 2018 PMID: 30179222
p.Lys966Asn	Missense	SRNS	Structurally modeled	Wang 2023 PMID: 37041680
c.1393dup (p.Ser465fs)	Frameshift	(ClinVar likely-pathogenic)	Loss of function	ClinVar

Variant classification landscape. ClinVar (queried Nov 2026) held 477 *NUP133* records but only **1 gene-specific likely-pathogenic small variant** (c.1393dup / p.Ser465fs), alongside 117 VUS and 39 (likely-)benign entries; the "Pathogenic" entries are large 1q copy-number gains spanning many genes rather than NUP133-specific. Consequently, most validated disease alleles reside in the **primary literature**, not ClinVar. This VUS-dominated landscape underscores the interpretive challenge in SRNS genetics (see Interpretative Challenges case, PMID: 42422724).

Allele frequency / population constraint. gnomAD v2.1.1 constraint for NUP133: **pLI = 0.00017** (not haploinsufficient), observed/expected LoF = 66/146.2 ≈ **0.45** (LOEUF ≈ 0.554), lof_z = 5.63, missense Z = 1.12. The low pLI indicates heterozygous loss-of-function is tolerated in the general population — consistent with recessive inheritance and healthy carriers — while the moderate oe_lof (~0.45) indicates partial depletion of biallelic LoF. This constraint profile is the molecular-genetic signature of an autosomal-recessive, carrier-tolerated disease gene.

Somatic vs germline. All disease variants are **germline**. No somatic involvement.

Functional consequences. Variants are **loss-of-function / hypomorphic**, destabilizing the Y-complex and weakening NUP133–NUP107 binding; no gain-of-function or dominant-negative mechanism is implicated.

Modifier genes. Not formally established; Y-complex paralogs (*NUP107*, *NUP85*, *NUP160*) are mechanistically related genes causing overlapping SRNS.

Epigenetic / chromosomal abnormalities. No disease-specific epigenetic mechanism is described. Large 1q copy-number gains encompassing NUP133 exist in ClinVar but are not the recessive point-mutation mechanism of NPHS18.

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5. Environmental Information

Not applicable as a cause. NPHS18 is a purely genetic Mendelian disorder. No environmental toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents cause or trigger the disease. Environmental factors are relevant only to the general management of nephrotic syndrome (e.g., sodium intake affecting edema) and to secondary complications (e.g., infection risk in the immunocompromised nephrotic state), not to disease etiology.

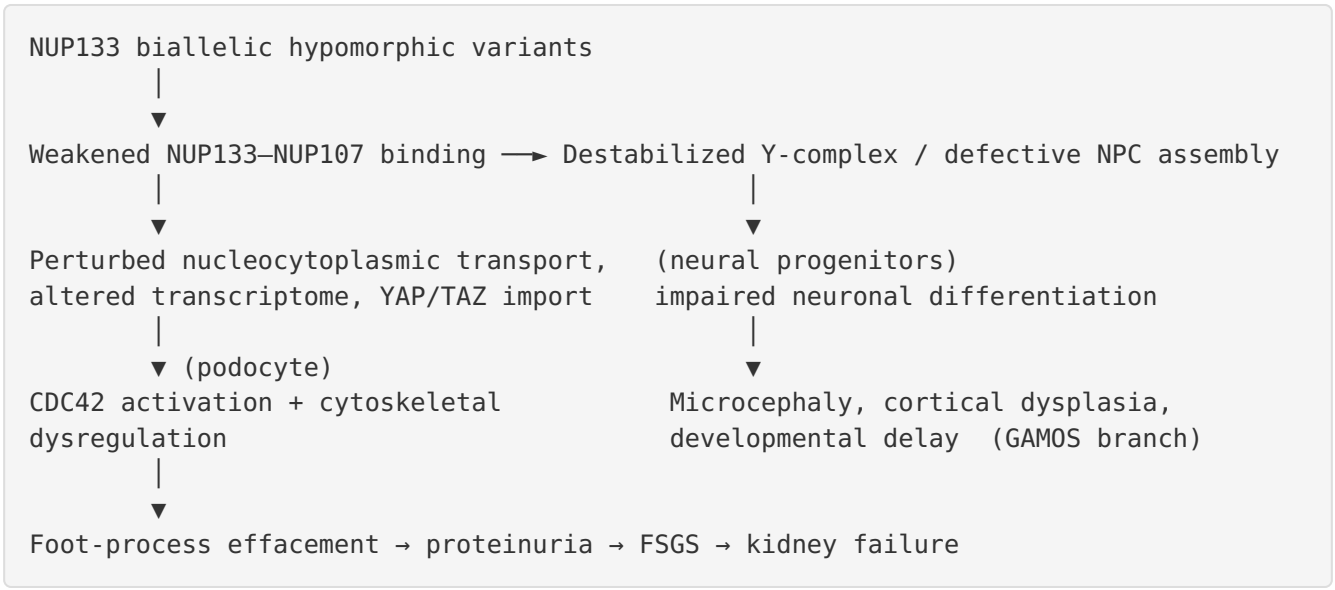
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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic *NUP133* variants** (splice/missense/frameshift) → **reduce functional NUP133 protein** and weaken its binding to the neighboring nucleoporin **NUP107** (demonstrated by co-IP; Braun 2018, Fujita 2018).
2. Impaired NUP133–NUP107 interaction → **destabilizes the Nup84/Y-complex of the NPC outer ring → defective nuclear pore complex assembly** (demonstrated in vitro; Rogg 2022, PMID: 35455939).
3. Defective NPC assembly → **perturbed nucleocytoplasmic transport and altered transcriptome composition**; NUP133 also participates in YAP/TAZ nuclear-import complexes (Ester 2023, PMID: 37565816) → altered nuclear signaling (partly inferred).
4. In podocytes specifically → **activation of CDC42** (Rho-family GTPase; a "key effector of SRNS pathogenesis") and **cytoskeletal dysregulation** (demonstrated by CRISPR/Cas9 KO in human podocytes; Braun 2018).
5. CDC42 activation + actin-cytoskeleton disruption → **podocyte injury and foot-process effacement** → breakdown of the glomerular filtration barrier → **nephrotic-range proteinuria** (demonstrated in zebrafish/Xenopus models; Braun 2018, Cianciolo Cosentino 2019 PMID: 30894603).
6. Sustained podocyte loss (podocytes cannot regenerate) → **focal segmental glomerulosclerosis (FSGS)** → progressive nephron loss → **kidney failure (median ~13 y)**.

Branch (neural, GAMOS): Steps 2–3 in neural progenitors → impaired neuroectodermal/neuronal differentiation and loss of nuclear-envelope integrity (Orniacki 2023 PMID: 37305998; Taniguchi 2025 PMID: 40205196) → **microcephaly, cortical malformation, developmental delay**.



Detail by category

- **Molecular pathways:** CDC42/Rho-GTPase signaling (central, downstream podocyte effector); Hippo–YAP/TAZ nuclear import (NUP133 is a component of YAP/TAZ complexes; NUP205 required for their nuclear import — Ester 2023). Upstream: nucleocytoplasmic transport / mRNA export machinery.
- **Cellular processes:** NPC assembly (GO:0051292), mRNA export from nucleus (GO:0006406), protein import into nucleus (GO:0006606), regulation of Rho-GTPase activity, cytoskeleton organization, podocyte foot-process maintenance; in GAMOS, neuronal differentiation and nuclear-envelope integrity.
- **Protein dysfunction:** Loss-of-function/hypomorphic — destabilized scaffold, weakened Y-complex contacts; not aggregation or GoF.
- **Immune involvement:** None primary (disease is immunosuppression-resistant, arguing against an immune-mediated podocytopathy). Secondary infection susceptibility arises from urinary immunoglobulin loss.
- **Tissue damage mechanism:** Podocyte injury → glomerulosclerosis/fibrosis; podocytes are terminally differentiated and non-regenerating, so loss is irreversible.
- **Molecular profiling:** In vitro human podocyte models show NUP133 controls NPC assembly, transcriptome composition, and cytoskeleton regulation (Rogg 2022). Single-cell/proteomic work on the related NUP160 KO mouse confirms Cdc42 dysregulation in podocytes (Qiao 2025, PMID: 40298220).

Upstream vs downstream: The NPC-scaffold/Y-complex defect is **upstream**; CDC42 activation, cytoskeletal collapse, foot-process effacement, and FSGS are **downstream**. Cell types: podocyte (CL:0000653, glomerular visceral epithelial cell), neurons/neural progenitors. Key GO biological processes: GO:0006406 (mRNA export), GO:0051292 (NPC assembly), GO:0001822 (kidney development), GO:0072006 (nephron development).

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7. Anatomical Structures Affected

- **Organ level — primary:** Kidney (UBERON:0002113), specifically the renal glomerulus (UBERON:0000074). Secondary involvement: whole urinary system; systemic effects of nephrotic syndrome. In GAMOS, the **brain / cerebral cortex** (UBERON:0000955) is primarily affected.
- **Body systems:** Renal/urinary (primary); nervous system (syndromic GAMOS); cardiovascular and hematologic (secondary — thrombosis, hyperlipidemia); immune (secondary infection risk).
- **Tissue/cell level:** Glomerular visceral epithelial cells — **podocytes (CL:0000653)** — are the central target; the glomerular basement membrane (UBERON:0002196) is secondarily disrupted. Neurons/neural progenitors in GAMOS.
- **Subcellular level: Nuclear pore complex (GO:0005643)** and **nuclear pore outer ring (GO:0031080)**; nuclear envelope; nucleus. NUP133 also localizes to kinetochore/centromere during mitosis.
- **Localization / lateralization:** Bilateral, diffuse glomerular involvement; microcephaly is symmetric/global.

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8. Temporal Development

- **Onset:** Childhood; median age at onset ~9 years for NUP133-associated SRNS (Yao 2025, PMID: 39331077). GAMOS forms present earlier (congenital microcephaly with early-onset nephrotic syndrome). Onset pattern is insidious/subacute with progressive proteinuria.
- **Progression:** Chronic and progressive. FSGS → declining GFR → **Stage 5 CKD/kidney failure at median ~13 years** for NUP133 (contrast: NUP93 ~3.7 y, NUP107 ~6.9 y). Progression rate is comparatively slower among nucleoporins for NUP133.

- **Disease course pattern:** Progressive (not relapsing-remitting); steroid-resistant, so no treatment-induced remission is expected. Duration: chronic/lifelong.
- **Patterns / critical periods:** No spontaneous remission. The key intervention window is **early genetic diagnosis** to redirect care away from ineffective immunosuppression and toward nephroprotection and transplant planning.

Nucleoporin gene	Median onset (y)	Median age at kidney failure (y)
NUP85	7	8.5
NUP93	3	3.7
NUP107	4.1	6.9
NUP133	**9**	**13**
NUP160	7	15
NUP205	2	7

(Yao 2025, PMID: 39331077)

9. Inheritance and Population

- **Inheritance pattern: Autosomal recessive.** Biallelic (homozygous or compound-heterozygous) variants required. Supported by gnomAD constraint (pLI ≈ 0; carriers healthy).
- **Penetrance / expressivity:** Penetrance in biallelic carriers appears high; expressivity is **variable**, spanning kidney-limited SRNS to syndromic GAMOS even for overlapping gene defects.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.
- **Consanguinity / founder effects:** Consanguinity strongly enriches recessive SRNS; NUP133 disease is frequently homozygous in consanguineous families (Fujita 2018; Azouaou 2026 PMID: 41612037). No specific founder allele is established.
- **Carrier frequency:** Not precisely defined; heterozygous LoF alleles are tolerated and present at low frequency in gnomAD.

- **Epidemiology:** No precise prevalence/incidence figures exist for NPHS18 specifically — it is an ultra-rare subset of monogenic SRNS. Context: ~30% of pediatric SRNS is monogenic (Nagano 2022, PMID: 36176665), rising to 56.1% with WGS in early-onset consanguineous cohorts (Soliman 2025, PMID: 40402239). NUP133 is one of several nucleoporin genes within this monogenic fraction.
- **Population demographics:** Reported across diverse populations; enriched in consanguineous populations (e.g., Middle Eastern, North African, South/Southeast Asian cohorts). No strong sex bias is expected for a recessive glomerular disorder. Age distribution: predominantly pediatric onset.

10. Diagnostics

- **Clinical/laboratory:** Nephrotic-range proteinuria (urine protein:creatinine, 24-h protein), serum albumin (hypoalbuminemia), serum lipids (hyperlipidemia). Clinical diagnosis of nephrotic syndrome = proteinuria + hypoalbuminemia + edema + hyperlipidemia.
- **Biopsy / pathology:** Kidney biopsy typically shows **FSGS** (light microscopy) with **diffuse foot-process effacement** on electron microscopy — present in ~89% of nucleoporin-SRNS biopsies (Yao 2025). Related nucleoporin studies show ultrastructural podocyte vacuolization, GBM changes, and ER dilation (Chen 2025, PMID: 39814977).
- **Genetic testing (definitive):** Molecular confirmation is essential. Recommended approach:
- **Targeted NGS gene panels** (SRNS/nephrotic-syndrome panels including NUP133 and paralogs) — most cost-effective, recommended first-line (Cheng 2024, PMID: 38654395; Yao 2025).
- **Whole-exome sequencing (WES)** — adds ~8.7% yield over panels.
- **Whole-genome sequencing (WGS)** — adds ~4.3% over WES; highest overall yield (56.1% in early-onset consanguineous cohorts).
- Single-gene testing and cDNA/minigene splice assays confirm candidate variants (e.g., c.3335-11T>A splice effect).
- CMA/karyotype/FISH/mtDNA/repeat-expansion testing: **not applicable** (point/splice recessive variants, not structural/mitochondrial/repeat disease).
- **Clinical criteria / differential diagnosis:** Differentiate from other monogenic SRNS/podocytopathies (NPHS1, NPHS2, PLCE1, WT1/Denys-Drash, LAMB2, TRPC6, COQ6, WDR73) and from immune-mediated primary FSGS/minimal change disease (which are steroid- or

rituximab-responsive). GAMOS differential includes other KEOPS-complex (OSGEP/TP53RK/TPRKB/LAGE3/GON7) and WDR73/WDR4/PRDM15 genes (Zhang 2025, PMID: 40533795).

- **Screening:** Cascade genetic testing of at-risk relatives; carrier testing in consanguineous families. No population newborn screening exists for this ultra-rare disorder.

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11. Outcome / Prognosis

- **Renal prognosis:** Poor without transplant. Progressive FSGS → **kidney failure at median ~13 years** for NUP133 (Yao 2025). Immunosuppression does not alter the natural history.
- **Survival/mortality:** No disease-specific survival statistics for NPHS18 alone; prognosis is governed by progression to kidney failure (manageable by dialysis/transplant) and, in GAMOS, by neurological severity. GAMOS carries higher early mortality due to brain involvement.
- **Morbidity/QoL:** Pediatric kidney failure, dialysis dependence, transplant, and (in syndromic cases) neurodevelopmental disability. Secondary complications: venous/arterial thrombosis, infection, cardiovascular risk (Ponticelli 2023, PMID: 37466816).
- **Post-transplant outcome:** Favorable relative to immune-mediated FSGS — genetic (podocyte-intrinsic) SRNS has **low recurrence** after transplantation (~12.5% for nucleoporin-SRNS, reported only for NUP93; Yao 2025), because the defect is not a circulating permeability factor.
- **Prognostic factors:** Genotype (which nucleoporin; NUP133 has comparatively later kidney failure), age at onset, presence of syndromic/neurological features, and degree of FSGS/fibrosis at biopsy.

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12. Treatment

Core principle: NUP133-SRNS is immunosuppression-resistant. "Most affected individuals are resistant to immunosuppressants" (Yao 2025, PMID: 39331077). Therapy is supportive/nephroprotective and, ultimately, kidney replacement.

Treatment	Class / NCIT	Rationale
ACE inhibitors / ARBs	RAAS blockade (NCIT C734 ACE inhibitor; Angiotensin Receptor Antagonist)	Antiproteinuric, nephroprotective
Loop diuretics + sodium restriction	Diuretic	Manage edema
Statins	HMG-CoA reductase inhibitor	Manage hyperlipidemia
Anticoagulation / thromboprophylaxis	Anticoagulant	Counter nephrotic hypercoagulability (urinary anticoagulant loss)
Vaccination / infection prevention	—	Counter urinary immunoglobulin loss
Kidney transplantation	NCIT C15366	Definitive therapy for kidney failure; low recurrence

- **Immunosuppression (steroids, CNIs, MMF, rituximab):** Generally **ineffective** in monogenic nucleoporin-SRNS; prolonged ineffective immunosuppression causes net harm (toxicity: e.g., CNI-induced pancreatitis, infections — see PMID: 42422724). Avoiding this is the main genotype-guided benefit of early diagnosis (Berke 2026, PMID: 42707999).
- **Advanced therapeutics (gene/RNA/cell therapy):** None approved or in trials for NUP133. Experimental mechanistic interest centers on the **CDC42 axis** downstream of NUP133 (Braun 2018).
- **Pharmacogenomics / personalized medicine:** The key personalized-medicine action is genotype-guided de-escalation of immunosuppression and transplant planning.
- **Supportive care:** Nutrition, blood-pressure control, cardiovascular risk management, management of CKD complications (Ponticelli 2023; Braun 2025 PMID: 40760331).

13. Prevention

- **Primary prevention:** None exists for a Mendelian recessive disorder.
- **Secondary/tertiary prevention:** Early genetic diagnosis to (a) stop ineffective/toxic immunosuppression, (b) institute nephroprotection (RAAS blockade), and (c) prevent

complications (thromboprophylaxis, infection prevention, cardiovascular risk management).

- **Genetic counseling:** Autosomal recessive — **25% recurrence risk per pregnancy** for carrier couples. Offer carrier testing of relatives and reproductive options: prenatal diagnosis and **preimplantation genetic testing (PGT)**. Consanguinity counseling is particularly important in high-consanguinity populations.
- **Immunization / public health / environmental interventions:** Not applicable to disease prevention (genetic etiology); standard vaccination is advisable given nephrotic infection risk.

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14. Other Species / Natural Disease

- **Conservation:** NUP133 is evolutionarily conserved from yeast to human; the *S. cerevisiae* Nup84–Nup133 Y-complex is structurally characterized (Nordeen 2020, PMID: 33247142).
- **Orthologs:** Mouse *Nup133* (NCBI Gene 234865; NCBITaxon:10090); zebrafish *nup133* (NCBITaxon:7955); *Xenopus* (NCBITaxon:8355); *S. cerevisiae NUP133* (NCBITaxon:4932).
- **Natural disease in other species: None catalogued** — no naturally occurring NUP133 disease is recorded in OMIA for companion animals or wildlife. No veterinary relevance or zoonotic potential (the disease is genetic, not transmissible).

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15. Model Organisms

Model	Type	Phenotype recapitulation	Source
Zebrafish *nup133* morphant	In vivo vertebrate	Proteinuria, podocyte foot-process effacement, kidney cysts, microcephaly, reduced neurons	Cianciolo Cosentino 2019 PMID: 30894603; Fujita 2018 PMID: 30427554
Xenopus nup133 morpholino	In vivo vertebrate	Disrupted glomerulogenesis; rescued by WT but not mutant mRNA	Braun 2018 PMID: 30179222
Human podocyte CRISPR/Cas9 KO	In vitro cellular	CDC42 activation; transcriptome/cytoskeleton dysregulation	Braun 2018; Rogg 2022 PMID: 35455939
Mouse ESC / neural progenitors (Nup133-deficient)	In vitro differentiation	Normal pluripotency; defective neuronal differentiation	Orniacki 2023 PMID: 37305998; Taniguchi 2025 PMID: 40205196
Podocyte-specific *Nup160* KO mouse (paralog)	In vivo mammalian	Proteinuria, foot-process fusion, Cdc42 dysregulation (validates pathway)	Qiao 2025 PMID: 40298220

- **Model limitations:** Constitutive *Nup133*-null mice are **not a viable disease model** — the gene is essential/embryonic-lethal. Conditional (podocyte-specific) and cellular models are therefore required. Zebrafish/Xenopus morphants model developmental renal and neural phenotypes well but are transient knockdowns.
- **Applications:** These systems established the CDC42 pathway (podocyte models), the neuronal-differentiation dependence (mESC), and the renal developmental phenotype (fish/frog), collectively supporting the tissue-specific-vulnerability model.
- **Resources:** MGI (mouse), ZFIN (zebrafish), Xenbase (Xenopus), SGD (yeast), Alliance of Genome Resources.

Mechanistic Model / Interpretation

The unifying insight of NPHS18 is **cell-type-specific vulnerability to a ubiquitous housekeeping defect**. NUP133 is a scaffold nucleoporin required for assembling every nuclear pore in every cell, yet biallelic hypomorphic variants produce a phenotype largely confined to podocytes (renal) and, when more severe, developing neurons (GAMOS). Three observations reconcile this paradox:

1. **The variants are hypomorphic, not null** — complete loss is embryonic-lethal (mouse); patients retain partial NUP133 function, placing stress on the most-vulnerable cell types.
2. **Podocytes and neurons are uniquely sensitive** — podocytes are terminally differentiated, non-regenerating cells maintaining an elaborate actin-based foot-process architecture; neurons depend on NUP133 for differentiation and nuclear-envelope integrity. mESC experiments show Nup133 is dispensable for pluripotency but essential for neuroectodermal/neuronal differentiation.
3. **A specific downstream effector (CDC42) links the nuclear-pore defect to podocyte injury** — providing a mechanistic "final common pathway" shared across outer-ring nucleoporin SRNS (NUP107/NUP85/NUP133 activate CDC42; NUP160 loss likewise dysregulates Cdc42).

This model positions the NPC-scaffold/Y-complex defect **upstream** and CDC42-driven cytoskeletal collapse, foot-process effacement, and FSGS **downstream**, with a GAMOS branch when neural progenitors are affected during development. Protein-structural context: NUP133 comprises an N-terminal β-propeller (harboring an amphipathic lipid-packing-sensor, ALPS, membrane-anchoring motif) and a C-terminal α-solenoid that binds NUP107; it recruits NUP107 and ELYS to membrane fenestrations during post-mitotic NPC reassembly (Nordeen 2020 PMID: 33247142; Bilir 2019 PMID: 30821042).

Evidence Base

PMID	Title (abbrev.)	Contribution
30179222	Mutations in multiple NPC components cause NS	**Landmark:** defines NUP133 as a causal SRNS gene; CDC42 activation mechanism; Xenopus rescue
30427554	Homozygous splicing mutation in NUP133 causes GAMOS	Syndromic spectrum; c.3335-11T>A; impaired NUP133–NUP107 binding; zebrafish neural+renal phenotype
39331077	Nucleoporin-associated SRNS (review)	Epidemiology (onset ~9 y, kidney failure ~13 y), 89% FSGS, immunosuppression resistance, extrarenal spectrum
35455939	NUP133 controls NPC assembly, transcriptome, cytoskeleton	Podocyte-specific pathomechanism
37565816	NUP205/YAP-TAZ in podocytes	NUP133 is a YAP/TAZ-complex component; Hippo-pathway link
37305998	Y-complex nucleoporins in neuronal progenitors	Tissue-specific differentiation dependence
40205196	Nuclear pores safeguard nuclear-envelope integrity	Neuronal differentiation dependence; GAMOS relevance
30894603	Moderate Nup133 deficiency → glomerular damage (zebrafish)	Renal phenotype recapitulation
40298220	Nup160 KO mouse dysregulates Cdc42	Validates CDC42 pathway in a mammalian in-vivo paralog model
33247142	Yeast Nup84–Nup133 structure, ALPS motif	Protein architecture/conservation
30821042	Nup133/Nup153 in NPC assembly	NUP133 recruits NUP107 and ELYS
37041680	NUP133 p.Lys966Asn structural modeling	Variant-level functional insight
36176665	Genetic characteristics of FSGS	~30% of pediatric SRNS is monogenic
40402239	WGS in early-onset SRNS	56.1% diagnostic yield in consanguineous families
38654395		Panel vs WES vs WGS yields/costs

PMID	Title (abbrev.)	Contribution
	Genetic testing modalities in SRNS	
37466816	NS pathophysiology and consequences	Hypercoagulability/thromboprophylaxis rationale

Limitations and Knowledge Gaps

1. **Rarity:** NPHS18 is known from a small number of families; precise prevalence/incidence, penetrance, and carrier frequency are undefined. No large natural-history cohort exists.
2. **Variant landscape:** ClinVar is dominated by VUS (117) with only one gene-specific likely-pathogenic small variant; most validated alleles are in primary literature. Variant interpretation remains a major clinical challenge.
3. **Genotype–phenotype correlation:** The determinants of the SRNS-vs-GAMOS spectrum (why some biallelic variants cause isolated kidney disease and others cause neuro-renal syndrome) are incompletely understood; modifier genes are unproven.
4. **CDC42 mechanism:** How NPC dysfunction mechanistically activates CDC42 in podocytes (direct vs indirect, transport-dependent vs transcriptional) is not fully resolved.
5. **No definitive mammalian NUP133 kidney model:** Constitutive knockout is lethal; a podocyte-specific *Nup133* conditional mouse (analogous to the existing *Nup160* model) is lacking.
6. **No disease-modifying therapy:** Treatment is entirely supportive; no gene/RNA/cell therapy or CDC42-targeted drug exists for patients.

Proposed Follow-up Experiments / Actions

1. **Generate a podocyte-specific conditional *Nup133* knockout mouse** (mirroring the *Nup160^podo−/−^* model) to definitively characterize the in-vivo renal pathomechanism and test the CDC42 hypothesis causally.

2. **Reclassify *NUP133* VUS** via systematic functional assays (minigene/splice assays, NUP107 co-IP binding, podocyte CDC42 activation) and ClinGen variant curation to reduce the VUS burden.
3. **Test CDC42-pathway inhibition** (e.g., CDC42/Rac inhibitors) in human podocyte and zebrafish models as a candidate targeted therapy for nucleoporin-SRNS.
4. **Build a NUP133 genotype–phenotype registry** linking specific alleles to SRNS-only vs GAMOS outcomes and to age at kidney failure, to improve prognostic counseling.
5. **Define carrier frequency and population distribution** using gnomAD and targeted consanguineous-cohort sequencing to inform reproductive counseling and cascade testing.
6. **Patient-derived iPSC podocyte/kidney-organoid models** carrying patient NUP133 variants to study transcriptome/cytoskeleton phenotypes and screen for rescuing compounds.
